

COMPANION
to

ChatGPT Teaches TikZ

Chapter 17: Graph Drawing (I)

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17.1 What This Companion Adds

Chapter 17 built the Petersen graph by placing vertices and then adding related families of edges. This Companion turns that example into a general graph drawing method: separate combinatorial structure, geometric placement, and visual style.

Design

Vertex names record incidence. Coordinates record placement. Styles record appearance. Keep these three roles separate.

17.2 A Graph-Drawing Vocabulary

Form or command	Purpose
<code>\coordinate (v1)</code>	Creates a named vertex position.
<code>(90:2)</code>	Places a point using polar coordinates.
<code>\foreach</code>	Repeats vertex or edge commands.
<code>(u)--(v)</code>	Draws an edge between named vertices.
<code>cycle</code>	Closes a cyclic edge sequence.
<code>circle</code>	Gives a vertex a circular mark.
<code>node</code>	Adds a visible vertex label.
<code>\tikzset</code>	Defines shared vertex and edge styles.

17.3 Three Independent Decisions

For every graph drawing, decide separately:

1. which vertices and edges form the graph;
2. where the vertices should appear; and
3. how vertices, edges, and labels should look.

Moving a vertex changes the drawing but not the graph. Adding an edge changes the graph. Changing a style changes only the appearance.

17.4 Placing Repeated Vertices

Polar coordinates place five outer and five inner vertices with one angle list.

```
\foreach \k/\angle in
  {1/90,2/18,3/-54,4/-126,5/162}
{
  \coordinate (O\k) at (\angle:2);
  \coordinate (I\k) at (\angle:.9);
}
```

The names O1 through O5 and I1 through I5 can now be used without repeating coordinates.

17.5 Drawing the Outer Cycle

```
\draw[very thick]
(O1)--(O2)--(O3)--(O4)--(O5)--cycle;
```

The order of the names records the five-cycle.

17.6 Drawing the Inner Star

```
\draw[very thick]
(I1)--(I3)--(I5)--(I2)--(I4)--cycle;
```

Connecting I1,I2,I3,I4,I5 in numerical order would create another pentagon, not the required star.

Common Mistake

When an edge pattern is wrong, inspect the sequence of vertex names before changing any coordinates.

17.7 Generating the Spokes

Each outer vertex is adjacent to the inner vertex with the same index.

```
\foreach \k in {1,...,5}  
  \draw (O\k)--(I\k);
```

The loop exposes the repeated combinatorial rule.

17.8 Drawing Vertices Last

Draw edges first and place the vertex marks over their endpoints.

```
\foreach \k in {1,...,5}  
  \filldraw[fill=white] (O\k) circle (2pt);  
  
\foreach \k in {1,...,5}  
  \filldraw (I\k) circle (2pt);
```

This prevents edge ends from showing through hollow vertex marks.

17.9 Separating Structure from Style

```
\tikzset{  
  vertex/.style={circle,draw,fill=white,inner sep=1.5pt},  
  edge/.style={thick}  
}
```

Once these styles exist, a global visual change does not require editing every vertex or edge.

17.10 Worked Project: The Petersen Graph

17.10.1 Stage 1: Define Placement

```
\foreach \k/\angle in
  {1/90,2/18,3/-54,4/-126,5/162}
{
  \coordinate (O\k) at (\angle:2);
  \coordinate (I\k) at (\angle:.9);
}
```

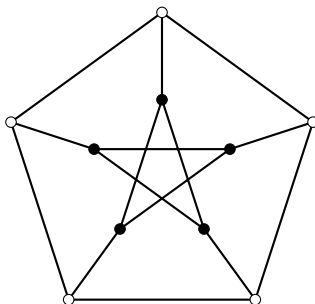
17.10.2 Stage 2: Encode the Three Edge Families

```
\draw (O1)--(O2)--(O3)--(O4)--(O5)--cycle;
\draw (I1)--(I3)--(I5)--(I2)--(I4)--cycle;
\foreach \k in {1,...,5} \draw (O\k)--(I\k);
```

17.10.3 The Complete Picture

```
\begin{tikzpicture}[scale=1.15]
  \tikzset{
    vertex/.style={circle,draw,fill=white,inner sep=1.4pt},
    edge/.style={thick}
  }
  \foreach \k/\angle in
    {1/90,2/18,3/-54,4/-126,5/162}
  {
    \coordinate (O\k) at (\angle:2);
    \coordinate (I\k) at (\angle:.9);
  }
  \draw[edge] (O1)--(O2)--(O3)--(O4)--(O5)--cycle;
  \draw[edge] (I1)--(I3)--(I5)--(I2)--(I4)--cycle;
  \foreach \k in {1,...,5}
    \draw[edge] (O\k)--(I\k);
  \foreach \k in {1,...,5}
  {
    \node[vertex] at (O\k) {};
    \node[vertex,fill=black] at (I\k) {};
  }
\end{tikzpicture}
```

See



Design

The coordinate loop records symmetry. The two explicit cycles and the spoke loop record the graph. The styles record appearance.

17.11 Checking the Combinatorics

The Petersen graph has ten vertices and fifteen edges. Every vertex has degree three:

Vertex type	Three incident edges
Outer O_k	Two outer-cycle edges and one spoke.
Inner I_k	Two inner-star edges and one spoke.

Count degrees from the edge rules, not from visual proximity.

17.12 A Small Graph from Scratch

The same method draws a square with one diagonal.

```
\coordinate (a) at (0,0);
\coordinate (b) at (2,0);
\coordinate (c) at (2,2);
\coordinate (d) at (0,2);
\draw (a)--(b)--(c)--(d)--cycle;
\draw (a)--(c);
\foreach \v in {a,b,c,d}
  \node[circle,draw,fill=white,inner sep=1.5pt] at (\v) {};
```

The diagonal changes the graph; moving `c` changes only its drawing.

17.13 Debugging Workshop

17.13.1 The Inner Shape Is a Pentagon

Use the star order I1,I3,I5,I2,I4, not numerical order.

17.13.2 Edges Cover the Vertex Centers

Draw the edges before the filled or hollow vertex marks.

17.13.3 Moving a Vertex Breaks Several Edges

Name the vertex once and use that name in every incident edge.

17.13.4 A Style Change Requires Many Edits

Create shared vertex and edge styles with `\tikzset`.

17.14 Exercises

1. Change the inner and outer radii without changing any edge.
2. Add visible labels to all ten vertices.
3. Give outer and inner vertices different shared styles.
4. Verify the degree of every Petersen vertex from the source.
5. Draw a five-cycle with a loop.
6. Draw the complete graph on four vertices.
7. Create a bipartite graph with two columns of vertices.
8. Redraw one graph with new coordinates but identical incidences.

17.15 Solutions

17.15.1 Exercises 1–4

Exercise 1 changes only the radii in the two polar coordinates.

```
\foreach \k in {1,...,5}  
  \node[above] at (0\k) {$0_{\k}$};
```

```
\tikzset{  
  outer/.style={circle,draw,fill=white},  
  inner/.style={circle,draw,fill=gray!35}  
}
```

Exercise 4 counts two cycle or star edges plus one spoke at each vertex.

17.15.2 Exercises 5–7

```
\foreach \k/\a in {1/90,2/18,3/-54,4/-126,5/162}  
  \coordinate (v\k) at (\a:1.5);  
\draw (v1)--(v2)--(v3)--(v4)--(v5)--cycle;
```

For Exercise 6, place four vertices and draw all six unordered pairs. For Exercise 7, name left and right vertex sets separately, then list or generate only the required cross-edges.

Exercise 8 changes coordinate definitions while leaving every edge command unchanged.

17.16 ChatGPT Workshop

Ask ChatGPT to preserve incidences while improving placement and source structure.

1. Separate the vertex placement, edge generation, and styling in this graph.
2. Replace repeated coordinates with named vertices and loops.
3. Check the degree of every vertex from the edge commands.
4. Diagnose why the intended inner star appears as a pentagon.
5. Change the layout without adding or deleting any edge.

17.17 Final Checklist

Before continuing to Chapter 18, check that you can

1. separate graph structure from geometric placement;
2. place repeated vertices with polar coordinates and loops;
3. encode cycles, stars, and repeated spokes;
4. draw edges before vertex marks;
5. define shared vertex and edge styles;
6. verify degrees from edge rules; and
7. change a layout without changing the graph.